L21 Solving Initial Value Problems

Section 7.5

Consider the IVP (with constant coefficients):

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{(n-1)} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = f(t) \qquad (1)$$

$$y(0) = y_0 \ , \ y'(0) = y_1 \ , \ \ldots \ , \ y^{(n-1)}(0) = y_{n-1}$$

Method of Laplace Transforms for Solving IVP

1. Take the Laplace transform of both sides of eq(1).

2. Use the properties of the Laplace transform and the initial conditions to reduce eq(1) to an algebraic equation in $Y(s)$.

3. Solve $Y(s)$.

4. Determine the inverse Laplace transform and the solution is $y(t) = \mathscr{L}^{-1}\{Y(s)\}$.

NOTE: You will need the formula

$$\mathscr{L}\{f^{(n)}(t)\} = s^n \mathscr{L}\{f(t)\} - s^{n-1} f(0) - s^{n-2} f'(0) -$$

$$\cdots - f^{(n-1)}(0).$$

Especially, when solving a second-order IVP, we use

$$\mathscr{L}\{y'(t)\} = sY(s) - y(0)$$

and

$$\mathscr{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0).$$

not IVP

$a_2y'' + a_1y' + a_0y = f(t)$

$y(0) = A$

$y'(0) = B$

ex. Solve: $y' - 3y = e^{2t}$, $y(0) = 1$ using Laplace transforms.

Apply $\mathcal{L}$ to both sides: $\mathcal{L}\{y' - 3y\} = \mathcal{L}\{e^{2t}\}$, $y(0) = 1$

$$(sY(s) - y(0)) - 3Y(s) = \frac{1}{s-2}$$

$$\Rightarrow \quad sY(s) - 1 - 3Y(s) = \frac{1}{s-2}$$

$$\Rightarrow \quad Y(s)(s-3) = \frac{1}{s-2} + 1$$

$$\Rightarrow \quad Y(s) = \frac{1}{(s-2)(s-3)} + \frac{1}{s-3}$$

$$\frac{1}{(s-2)(s-3)} \overset{PFD}{=} \frac{A}{s-2} + \frac{B}{s-3} \Rightarrow 1 = A(s-3) + B(s-2)$$

$s = 2: \ A = -1$

$s = 3: \ B = 1$

$$\Rightarrow Y(s) = \left[\frac{-1}{s-2} + \frac{1}{s-3}\right] + \frac{1}{s-3} \Rightarrow \boxed{y(t) = \mathcal{L}^{-1}\{Y(s)\} = -e^{2t} + 2e^{3t}}$$

ex. Solve: $y'' - 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = -1$ using Laplace Transforms.

$$\Rightarrow \mathcal{L}\{y'' - 6y' + 9y\} = \mathcal{L}\{0\}$$

$$\Rightarrow \left(s^2Y(s) - sy(0) - y'(0)\right) - 6\left(sY(s) - y(0)\right) + 9Y(s) = 0$$

$$\Rightarrow s^2Y(s) - s(1) - (-1) - 6sY(s) + 6(1) + 9Y(s) = 0$$

$$\Rightarrow Y(s)\left(s^2 - 6s + 9\right) - s + 7 = 0$$

$$\Rightarrow Y(s) = \frac{s-7}{s^2-6s+9} = \frac{s-7}{(s-3)^2}$$

$s - 3 - 4$

$$= \frac{(s-3)-4}{(s-3)^2} = \frac{s-3}{(s-3)^2} - \frac{4}{(s-3)^2} = \frac{1}{s-3} - \frac{4}{(s-3)^2}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{1}{s-3} - \frac{4}{(s-3)^2}\right\} = e^{3t} - 4e^{3t}t$$

ex. Solve: $y'' + 6y' + 5y = 12e^t$, $y(0) = -1$, $y'(0) = 7$ using Laplace transforms.

$$\left(s^2 Y(s) - s\overset{-1}{y(0)} - \overset{7}{y'(0)} \right) + 6\left(sY(s) - \overset{-1}{y(0)} \right) + 5Y(s) = \frac{12}{s-1}$$

$$\Rightarrow (s^2 + 6s + 5)Y(s) + s - 1 = \frac{12}{s-1}$$

$$\Rightarrow (s+5)(s+1)Y(s) = \frac{12}{s-1} - s + 1$$

$$\Rightarrow Y(s) = \frac{12}{(s+5)(s-1)(s+1)} + \frac{1-s}{(s+5)(s+1)}$$

$$\overset{PFD}{=} \left(\frac{1/2}{s+5} + \frac{1}{s-1} - \frac{3/2}{s+1} \right) + \left(\frac{-3/2}{s+5} + \frac{1/2}{s+1} \right)$$

$$\frac{12}{(s+5)(s-1)(s+1)} = \frac{A}{s+5} + \frac{B}{s-1} + \frac{C}{s+1}$$

$$\Rightarrow 12 = A(s-1)(s+1) + B(s+5)(s+1) + C(s+5)(s-1)$$

$$\Rightarrow A = \frac{1}{2}, \quad B = 1, \quad C = -\frac{3}{2}$$

$$\frac{1-s}{(s+5)(s+1)} = \frac{A}{s+5} + \frac{B}{s+1}$$

$$\Rightarrow 1 - s = A(s+1) + B(s+5)$$

$$\Rightarrow A = -\frac{3}{2}, \quad B = \frac{1}{2}$$

$$\Rightarrow Y(s) = \frac{-1}{s+5} + \frac{1}{s-1} - \frac{1}{s+1}$$

$$\Rightarrow y(t) = -e^{-5t} + e^{t} - e^{-t}$$

$$\mathcal{L}\{t^2e^{3t}\} = \frac{2}{s^3}\Big|_{s\to s-3}$$

ex. Solve: $y'' - 6y' + 9y = t^2e^{3t}$, $y(0) = 2$, $y'(0) = 6$ using Laplace transforms.

$$(s^2Y(s) - s\cancel{y(0)}^{2} - \cancel{y'(0)}^{6}) - 6(sY(s) - \cancel{y(0)}^{2}) + 9Y(s) = \frac{2}{(s-3)^3}$$

$$\Rightarrow (s^2 - 6s + 9)Y(s) - 2s + 6 = \frac{2}{(s-3)^3}$$

$$\Rightarrow \quad Y(s)(s-3)^2 = \frac{2}{(s-3)^3} + 2(s-3)$$

$$\Rightarrow \quad Y(s) = \frac{2}{(s-3)^5} + \frac{2}{(s-3)} = \frac{2}{4!}\frac{4!}{(s-3)^5} + 2\frac{1}{(s-3)}$$

$$\Rightarrow y(t) = \frac{2}{24}e^{3t}t^4 + 2e^{3t}$$

$$= e^{3t}\left(\frac{1}{12}t^4 + 2\right)$$

ex. Solve: $y'' + 4y' + 6y = 1 + e^{-t}$, $y(0) = 0$, $y'(0) = 0$ using Laplace transforms.

$$\mathcal{L}\{y'' + 4y' + 6y\} = \mathcal{L}\{1 + e^{-t}\}$$

$$\Rightarrow \left(s^2 Y(s) - \cancel{s y(0)}^{0} - \cancel{y'(0)}^{0}\right) + 4\left(sY(s) - \cancel{y(0)}^{0}\right) + 6Y(s) = \frac{1}{s} + \frac{1}{s+1}$$

$$\Rightarrow s^2 Y(s) + 4sY(s) + 6Y(s) = \frac{1}{s} + \frac{1}{s+1}$$

$$\Rightarrow Y(s)(s^2 + 4s + 6) = \frac{1}{s} + \frac{1}{s+1}$$

$$\Rightarrow Y(s) = \frac{1}{s(s^2 + 4s + 6)} + \frac{1}{(s+1)(s^2 + 4s + 6)}$$

$+4-4$

$$= \frac{1}{s((s+2)^2 + 2)} + \frac{1}{(s+1)((s+2)^2 + 2)}$$

$(s+2)^2 + 2$

$\alpha = -2$ $\beta = \sqrt{2}$

$e^{\alpha t}\begin{pmatrix}\cos(\beta t) \\ \sin(\beta t)\end{pmatrix}$

$$\frac{1}{s((s+2)^2+2)} = \frac{A}{s} + \frac{B(s+2) + C\cdot\sqrt{2}}{(s+2)^2 + 2} \Rightarrow 1 = A[(s+2)^2 + 2] + [B(s+2) + \sqrt{2}C]s$$

$s = 0$: $1 = A[6] \Rightarrow A = \frac{1}{6}$, $s = -2$: $1 = \frac{1}{6}[2] + \sqrt{2}\cdot C(-2)$

$\Rightarrow -\frac{1}{\sqrt{2}\cdot 3} = C$

$s = 1$: $1 = \frac{1}{6}(11) + [3B + (-\frac{1}{3})](1) \Rightarrow B = -\frac{1}{6}$

$$\frac{1}{(s+1)((s+2)^2+2)} = \frac{A}{s+1} + \frac{B(s+2) + \sqrt{2}C}{(s+2)^2 + 2} \Rightarrow 1 = A[(s+2)^2 + 2] + [B(s+2) + \sqrt{2}C](s+1)$$

$s = -1$: $1 = 3A \Rightarrow A = \frac{1}{3}$, $s = -2$: $1 = \frac{2}{3} - \sqrt{2}C \Rightarrow C = -\frac{1}{3\sqrt{2}}$

$s = 0$: $1 = \frac{6}{3} + [2B - \frac{1}{3}](1) \Rightarrow B = \frac{1}{6}$

$$Y(s)=\left[\frac{1/6}{s}+\frac{-\frac{1}{6}(s+2)-\frac{1}{\sqrt{2}\cdot 3}\cdot\sqrt{2}}{(s+2)^2+2}\right]+\left[\frac{1/3}{s+1}+\frac{\frac{1}{6}(s+2)-\frac{1}{\sqrt{2}\cdot 3}\sqrt{2}}{(s+2)^2+2}\right]$$

The initial conditions are not given at $t=0$:

<u>ex.</u> Solve $w''-2w'+w=6t-2$ $w(-1)=3$ $w'(-1)=7$

Use the substitution $y(t)=w(t-1)$ to move the initial conditions to $t=0$.

$$\boxed{y(t)=\frac{1}{6}-\cancel{\frac{1}{6}e^{-2t}\cos(\sqrt{2}t)}-\frac{1}{\sqrt{2}\cdot 3}e^{-2t}\sin(\sqrt{2}t)+\frac{1}{3}e^{-t}+\cancel{\frac{1}{6}e^{-2t}\cos(\sqrt{2}t)}-\frac{1}{\sqrt{2}\cdot 3}e^{-2t}\sin(\sqrt{2}t)}$$

New IVP at $t=0$:

$$w''-2w'+w=6t-2\ ,\quad w(-1)=3,\ w'(-1)=7$$

set $v(t+1)=w(t)$, $t=-1$, $v(0)$

$\Rightarrow v(t)=w(t-1)$ $v(0)=3$, $v'(0)=7$

$$v''-2v'+v=6(t-1)-2\ ,\quad v(0)=3,\ v'(0)=7$$

$6t-8$

$$\Rightarrow (s^2V(s)-sv(0)-v'(0))-2(sV(s)-v(0))+V(s)=\frac{6}{s^2}-\frac{8}{s}$$

$$\Rightarrow V(s)(s^2-2s+1)-3s-7+6=\frac{6}{s^2}-\frac{8}{s}$$

$$\Rightarrow V(s)(s-1)^2=\frac{6}{s^2}-\frac{8}{s}+3s+1=\frac{6-8s+3s^3+s^2}{s^2}$$

$$\Rightarrow V(s)=\frac{6-8s+3s^3+s^2}{s^2(s-1)^2}=\frac{A}{s}+\frac{B}{s^2}+\frac{C}{(s-1)}+\frac{D}{(s-1)^2}$$

$$=\frac{6}{s^2}+\frac{4}{s}-\frac{1}{s-1}+\frac{2}{(s-1)^2}\quad v(t)=$$

$$=6t+4-e^t+2te^t$$

$v(t)=w(t-1)$

$$\Rightarrow \boxed{w(t)=v(t+1)=6(t+1)+4-e^{t+1}+2(t+1)e^{t+1}}$$

Solving linear equations whose coefficients are polynomials of t:

Apply the property

$$\mathscr{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

If $n = 1$ and $f(t) = y'(t)$, then

$$\mathscr{L}\{ty'(t)\} = -\frac{d}{ds}\mathscr{L}\{y'\} = -\frac{d}{ds}(sY(s) - y(0))$$

$$= -[Y(s) + sY'(s)]$$

$$= -sY'(s) - Y(s)$$

If $n = 1$ and $f(t) = y''(t)$, then

$$\mathscr{L}\{ty''(t)\} = -\frac{d}{ds}\mathscr{L}\{y''\} = -\frac{d}{ds}[s^2Y(s) - sy(0) - y'(0)]$$

$$= -[2sY(s) + s^2Y'(s) - y(0)]$$

$$= -s^2Y'(s) - 2sY(s) + y(0)$$

The method of Laplace transform will convert the DEQ into a linear differential equation in $Y(s)$.

$\lim_{s\to\infty} Y(s) = 0$

NOTE: You may need the theorem: $\lim_{s\to\infty} \mathscr{L}\{f(t)\} = 0$

<u>ex.</u> Solve: $y'' + ty' - y = 0$, $y(0) = 0$, $y'(0) = 3$

$$(s^2Y(s) - s\cancel{y(0)}^{\,0} - \underbrace{y'(0)}_{-3}) + [-sY'(s) - Y(s)] - Y(s) = 0$$

$$-sY'(s) + (s^2-2)Y(s) - 3 = 0 \qquad \text{1st-order linear}$$

$$\Rightarrow Y'(s) + \frac{s^2-2}{-s}Y(s) = -\frac{3}{s}$$

$$\mu(s) = e^{\int -\frac{s^2-2}{s}ds} = e^{\int -s + \frac{2}{s}ds} = e^{-\frac{1}{2}s^2 + 2\ln|s|} = s^2e^{-\frac{1}{2}s^2}$$

so $$Y(s) = \frac{1}{\mu}\left[\int \mu Q + C\right] = \frac{1}{s^2e^{-\frac{1}{2}s^2}}\left[\int s^2e^{-\frac{1}{2}s^2}\cdot -\frac{3}{s}ds + C\right]$$

$$\Rightarrow Y(s) = \frac{1}{s^2e^{-\frac{1}{2}s^2}}\left[\int -3se^{-\frac{1}{2}s^2}ds + C\right]$$

$u = -\frac{1}{2}s^2$

$du = -s\,ds$

$\int 3e^u$

$$= \frac{1}{s^2e^{-\frac{1}{2}s^2}}\left[3e^{-\frac{1}{2}s^2} + C\right]$$

$$\Rightarrow Y(s) = \frac{3}{s^2} + \cancel{\frac{C}{s^2}e^{1/2s^2}} \rightarrow \lim_{s\to\infty} Y(s) = 0 + \to \infty$$

$\to$ DNE

must mean $\underline{\underline{C = 0}}$.

$$\Rightarrow Y(s) = \frac{3}{s^2}$$

$$\Rightarrow y(t) = 3t$$

Extra / Lecture 29

$v(0) = B$

Consider $my'' + ky = 0$, $y(0) = A$, $y'(0) = B$

set $v = y'$, $v' = y''$ $\rightarrow$ $y'' = -\frac{k}{m}y \Rightarrow v' = -\frac{k}{m}y$

$$\Rightarrow \boxed{\begin{cases} y' = v \\ v' = -\frac{k}{m}y \end{cases}}$$

$$\Rightarrow \frac{dv}{dy} = \frac{dv/dt}{dy/dt} = \frac{-\frac{k}{m}y}{v}$$ phase-plane equation

$$\Rightarrow mv\,dv = -ky\,dy \Rightarrow \frac{1}{2}mv^2 + \frac{1}{2}ky^2 = C$$

ellipse

Consider
$$\begin{cases} y' = v, & y(0) = A \\ v' = -\frac{k}{m}y, & v(0) = B \end{cases}$$

use Laplace Transform to convert:

$$\mathcal{L}\{y' = v\} \Rightarrow sY(s) - y(0) = V(s)$$

$$\mathcal{L}\{v' = -\frac{k}{m}y\} \Rightarrow sV(s) - v(0) = -\frac{k}{m}Y(s)$$

$$\Rightarrow \begin{cases} sY(s) - V(s) = A \leftarrow \cdot s & (1) \\ \frac{k}{m}Y(s) + sV(s) = B & (2) \end{cases}$$

$$\Rightarrow s(1) + (2): \quad s^2Y(s) + \frac{k}{m}Y(s) = As + B$$

$$\Rightarrow Y(s)\left(s^2 + \frac{k}{m}\right) = As + B$$

$$\Rightarrow Y(s) = \frac{As + B}{s^2 + \frac{k}{m}}$$

$$\Rightarrow \quad Y(s) = \frac{As}{s^2 + \frac{k}{m}} + B \cdot \sqrt{\frac{m}{k}} \frac{\sqrt{\frac{k}{m}}}{s^2 + \frac{k}{m}}$$

$b = \sqrt{\frac{k}{m}}$

$$\Rightarrow \quad y(t) = A\cos\left(\sqrt{\tfrac{k}{m}}\, t\right) + B\sqrt{\tfrac{m}{k}}\, \sin\left(\sqrt{\tfrac{k}{m}}\, t\right)$$

$$\Rightarrow \quad v(t) = -A\sqrt{\tfrac{k}{m}}\sin\left(\sqrt{\tfrac{k}{m}}\, t\right) + B\cancel{\sqrt{\tfrac{m}{k}}} \cdot \cancel{\sqrt{\tfrac{k}{m}}}\cos\left(\sqrt{\tfrac{k}{m}}\, t\right)$$

$$= B\cos\left(\sqrt{\tfrac{k}{m}}\, t\right) - A\sqrt{\tfrac{k}{m}}\sin\left(\sqrt{\tfrac{k}{m}}\, t\right)$$

The trajectory in the phase-plane is given by $(y(t), v(t))$.

Consider $my'' + by' + ky = 0$, $y(0) = A$, $y'(0) = B$ $v(0)$

set $v = y' \Rightarrow y'' = -\frac{k}{m}y - \frac{b}{m}y'$

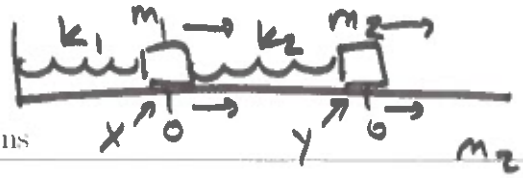
$$\Rightarrow \begin{cases} y' = v & , \ y(0) = A \\ v' = -\frac{k}{m}y - \frac{b}{m}v & , \ v(0) = B \end{cases}$$

underdamped $b^2 - 4mk < 0$

$$\Rightarrow \quad y(t) = A\underbrace{e^{-\frac{b}{2m}t}}_{\text{decay}} \cos\left(\sqrt{\frac{4mk - b^2}{4m^2}}\, t\right)$$

$$+ \left(B + A\left(\frac{2k - b}{2m}\right)\right)\left(\frac{2m}{\sqrt{4mk - b^2}}\right) e^{-\frac{b}{2m}t} \sin\left(\sqrt{\frac{4mk - b^2}{4m^2}}\, t\right)$$

$m_1 x'' = -k_1 x + (y-x)k_2$



no damping

$m_2 y'' = -(y-x)k_2$

Example: An object with mass m_1 is attached to a wall by a spring with another object with mass m_2 attached to the first by another spring. This is called a coupled mass-spring system. If the spring constant for the spring between the wall and first object is k_1 and for the other spring it is k_2, and the displacement from equilibrium of the first object is x while the displacement form equilibrium for the second object is y, then the equations of motion (assuming no friction force) of the two objects satisfies

$$\begin{cases} m_1 x'' + (k_1 + k_2)x - k_2 y = 0 \\ m_2 y'' + k_2 y - k_2 x = 0 \end{cases}$$

Suppose $m_1 = 2$, $m_2 = 1$, $k_1 = 4$, and $k_2 = 2$. If the two objects are initially displaced 3 units to the right, solve the system.

$$\Rightarrow \mathcal{L}\left(\begin{cases} 2x'' + 6x - 2y = 0 \\ y'' + 2y - 2x = 0 \end{cases}\right)$$

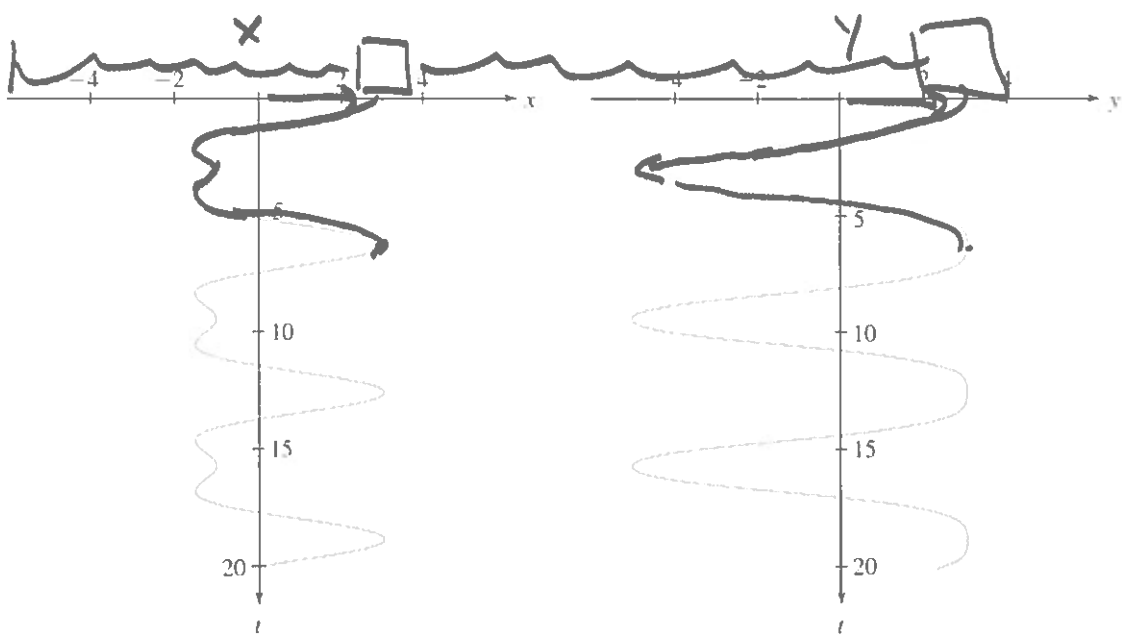
$$\Rightarrow \begin{cases} 2(s^2X(s) - s\cancel{x(0)}^{3} - \cancel{x'(0)}^{0}) + 6X(s) - 2Y(s) = 0 \\ s^2Y(s) - s\cancel{y(0)}^{3} - \cancel{y'(0)}^{0} + 2Y(s) - 2X(s) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} 2s^2X(s) - 6s + 6X(s) - 2Y(s) = 0 \\ s^2Y(s) - 3s + 2Y(s) - 2X(s) = 0 \end{cases}$$

$$\Rightarrow (s^2+3)X(s) - Y(s) = 3s$$
$$-2X(s) + (s^2+2)Y(s) = 3s$$

$$\Rightarrow X(s) = \frac{2s}{s^2+1} + \frac{s}{s^2+4} \Rightarrow x(t) = 2\cos(t) + \cos(2t)$$

$$\Rightarrow y(t) = 4\cos(t) - \cos(2t)$$

$$\Rightarrow \begin{cases} x(t) = 2\cos(t) + \cos(2t) \\ y(t) = 4\cos(t) - \cos(2t) \end{cases}$$

frequency 1 & 2 normal modes.